RESEARCH DIRECTIONS AND OPEN PROBLEMS AROUND CONDENSATION

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Contents

This is a survey of various ways that I'd like to see work on the theory of condensation develop. Condensation is a mathematical theory dealing with the organization of descriptions of the world into conceptual parts; some of the existing work on it is presented in the paper Eisenstat (2025). This document will draw on the definitions of *random variable models* and *latent variable models*, the notation for indexing subfamilies of variables in such models, and the *objectivity theorem*, Theorem 6.8, in that paper. Section 3, Ideas, of the paper gives an overview of all of this. For more context on condensation, readers can refer to LessWrong posts including Demski (2025), Demski (2026), Gillen and Chiang (2026), and Kirchner (2026). The sections on the different research directions are moderately interdependent, since the concepts of almost perfect condensation, introduced in Section §1, and Kolmogorov (or algorithmic-information) condensation, introduced in Section §2, are used in other sections.

I'd encourage those making a serious effort on any of these problem to contact me for further thoughts and coordination.

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1. Existence of condensations

For the theory of condensation to be useful, we'd like hypotheses that we can assume about latent variable models that are general enough so that such latent variable models exist in cases of interesting random variable models, but narrow enough to imply objectivity-like conclusions. We'll refer to hypotheses of this kind as *condensation properties.* To know that such properties apply to interesting data, we'd like to come up with interesting random variable models satisfying them. (We can also look for interesting *string models*, which will be the analogue of random variable models when we introduce Kolmogorov condensation in Section §2.) Work in these directions will also affect the problems discussed in later sections, which almost all either need to assume the existence of a latent variable model (or a *latent string model* in the Kolmogorov case) with good condensation properties, or else ask for such latent variable models in some particular context.

1.1. Almost perfect condensation.

One strong condensation property is almost perfect condensation. I've defined this somewhat differently in different places, and there isn't yet a good reference for this. We'll look at a definition here, and we'll discuss informally the consequences that follow from and motivate the definition, without giving a theorem statement or proof sketch.

Definition 1. Suppose that $(X_i)_{i\in I}$ are the random variables of a random variable model, and that we have an associated latent variable model $\mathcal{Y}$, with latent variables $(Y_A)_{A\in\mathcal{J}}$, where $\mathcal{J}\subseteq\mathcal{P}^+I$. Then, $\mathcal{Y}$ is an (m,r,ε)-*almost perfect condensation* if (1—size) $|\mathcal{J}|\leq m$, (2—reconstruction) for all $A\in\mathcal{J}$ and all $B\in\mathcal{P}^+I$, if $|A\cap B|>r$, then $H\left(Y_{\supseteq A}\mid X_B\right)\leq\varepsilon$, and (3—Markov) for all $\subseteq$-upward closed sets $\mathcal{F},\mathcal{G}\subseteq J$, we have $I\left(Y_{\mathcal{F}};Y_{\mathcal{G}}\mid Y_{\mathcal{F}\cap\mathcal{G}}\right)\leq\varepsilon$. Further, we say that $\mathcal{Y}$ is r-*well-separated* if (4) for all $A,B\in\mathcal{J}$, if $|A|>r$ then at least one of the sets $A-B$ and $B-A$ has cardinality greater than $2r$.

Here, we should have an objectivity theorem as follows. If $\mathcal{Y}$ and $\mathcal{Z}$ are two well-separated almost perfect condensations of the same string model, then this should induce a bijection between the latents of $\mathcal{Y}$ and $\mathcal{Z}$ with good objectivity-style mutual determination. More generally, if $\mathcal{Y}$ and $\mathcal{Z}$ are both almost perfect condensations, and $\mathcal{Y}$ is well-separated, then we get a partition of the latents of $\mathcal{Z}$ into some sets, and these sets admit a bijection with the latents of $\mathcal{Y}$ with similar properties to the previous case. Finally, if we only assume that $\mathcal{Y}$ and $\mathcal{Z}$ are almost perfect condensations, without any well-separatedness assumption, we can still conclude that each latent of $\mathcal{Y}$ is determined by some small set of latents of $\mathcal{Z}$, and vice versa.

We can construct toy examples of almost-perfect condensations. For example, in the Shannon information setting, we can consider the following example, where the latents are the biases of dice, and the given variables are observations that give us some information about these dice. That is, we will have a set of latents $(Y_j)_{j\in\widetilde{J}}$, each of which is valued in the space of probability measures on $\{1,\ldots,6\}$, and which are jointly independent. The idea here is that the bias of each die is itself random (we can think of drawing it from a bag of biased dice), but that particular die, with that bias, is rolled multiple times, giving us information about the bias. Now, for the construction, fix some sets $I,\widetilde{J}$, and a relation $\triangleright\subseteq\widetilde{J}\times I$. Construct a joint distribution on $(Y_j)_{j\in\widetilde{J}}$ as described. Define $(X_i)_{i\in I}$ by making a roll $R_{j,i}$

of die j, and setting

$$X_i = \sum_{j \triangleright i} R_{j,i}.$$

We need further latents to account for the idiosyncratic information in X_i, so set $J = \widetilde{J} \sqcup I$ and $Y_i = X_i$, extending $\triangleright$ to $J \times I$ by setting also $i \triangleright i$. We can check that for appropriate parameters, $\left((Y_j)_{j \in J}, \triangleright\right)$ is an almost perfect condensation of $(X_i)_{i \in I}$. We want $|I|$ sufficiently large relative to $\left|\widetilde{J}\right|$; not too many $j \in \widetilde{J}$ such that $j \triangleright i$ for each fixed $i \in I$ (e.g., we could choose $\triangleright$ such that there are exactly two such j for each i, that is, each observation is the sum of two die rolls); and $\triangleright$ sufficiently evenly distributed, avoiding properties like two different $j \in \widetilde{J}$ contributing to the same, or nearly the same, set of $i \in I$.

It would be helpful to have examples of almost perfect condensation that are a bit less artificial, and better demonstrate its relevance to constructing interesting latents. This could be in either the Shannon entropy or the Kolmogorov complexity setting. It would also be good if there were better ways to think about the rather long definition given, in which many somewhat arbitrary choices were made. For example, in definition 1, many things that we want to be small are individually bounded by the same ε. Any bounds would give some kind of objectivity theorem. Is there a better way of thinking about adapting these upper bounds to the different quantities bounded? This might be a situation where the bounds should be adapted to particular latent variable models being studied. But even then, there could be some general insight to be learned from looking at interesting such examples.

1.2. **Beyond almost perfect condensation.** Almost perfect condensation has been constructed so as to derive reasonably strong and simple bounds from the objectivity theorem, and more generally aid in the interpretation of its conclusions. However, there may be many other cases where the objectivity theorem can tell us that some latent model is approximately unique enough to be interesting. We will look at some examples and discuss why almost perfect condensation will be too strict to allow the sort of analysis that we'd want, and argue that there's reasonable hope for other ideas here. But first, we'll review the objectivity theorem so that we can see how it applies here. We'll work in the Shannon setting for simplicity, but these problems also exist in the Kolmogorov setting.

The general idea of the objectivity theorem is as follows. Let's say $\mathcal{X}$ is a random variable model with latent variables $(X_i)_{i \in I}$, and $\mathcal{Y}$ and $\mathcal{Z}$ are two associated latent variable models with latent variables $(Y_A)_{A \in \mathcal{J}}$ and $(Z_B)_{B \in \mathcal{K}}$, respectively, where $\mathcal{J}, \mathcal{K} \subseteq \mathcal{P}^+ I$. The objectivity theorem says that for any family $\mathcal{F} \subseteq \mathcal{P}^+ I$, we have

$$H\left(Y_{\supseteq A} \mid Z_{\mathcal{F}^\circ}\right) \leq \sum_{B \in \mathcal{F}} H\left(Y_{\supseteq A} \mid X_B\right) + \sum_{v \in N} I\left(Z_{\mathcal{L}(v)}; Z_{\mathcal{R}(v)} \mid Z_{\mathcal{I}(v)}\right).$$

Now, suppose that $\mathcal{Y}$ and $\mathcal{Z}$ are (m, r, ε)-almost perfect condensations; they satisfy the three conditions in definition 1. We can use this to pick $\mathcal{F}$ well in order to simultaneously control $\mathcal{F}^\circ$ and the right-hand side of the inequality. Using condition (1) for $\mathcal{Z}$, we know that $|\mathcal{K}| \leq m$. If we want to control $Z_{\mathcal{F}^\circ}$, we really only need to control $\mathcal{F}^\circ \cap \mathcal{K}$, and if we remove elements from $\mathcal{K}$ one at a time, this can take at most m-many steps. Recalling the definition

$$\mathcal{F}^\circ = \left\{C \in \mathcal{P}^+ I : \forall B \in \mathcal{F}.\, B \cap C \neq \emptyset\right\},$$

we can see that we can achieve any desired set $\mathcal{F}^\circ \cap \mathcal{K}$ with a set $\mathcal{F}$ of size at most m. That is, suppose that we want to bound $H\left(Y_{\supseteq A} \mid Z_{\mathcal{G}}\right)$ for some desired upward-closed set $\mathcal{G} \subseteq \mathcal{P}^+ I$. We can let

$$\mathcal{F} = \{I - B : B \in \mathcal{K} - \mathcal{G}\},$$

which will have size at most m, and will satisfy $\mathcal{G} \cap \mathcal{K} = \mathcal{F}^\circ \cap \mathcal{K}$.

By the definition of N in the objectivity theorem, we have $|N| = |\mathcal{F}| - 1$. Further, since $\mathcal{Z}$ satisfies the approximate Markov condition, condition (3) from the definition of almost perfect condensation, each of the terms $I\left(Z_{\mathcal{L}(v)}; Z_{\mathcal{R}(v)} \mid Z_{\mathcal{I}(v)}\right)$ in the sum over N is at most ε. Thus, this inequality implies the simpler statement

$$H\left(Y_{\supseteq A} \mid Z_{\mathcal{G}}\right) \leq \sum_{B \in \mathcal{F}} H\left(Y_{\supseteq A} \mid X_B\right) + m\varepsilon.$$

From this, we can motivate condition (2) in the definition of almost perfect condensation. Each set B on the right-hand-side sum is the complement of a set in $\mathcal{K}$ that we want to ensure is absent from $\mathcal{G}$. Further, we want $Y_{\supseteq A}$ to be reconstructable from—to have low conditional entropy given—Z_B. So, one simple condition to impose is that the intersection of A and B is sufficiently large. For a fixed r, we can take $\mathcal{G}$ to be the collections of sets C such that $|A - C| \leq r$, which we can regard as an approximation of the upward cone of A. This definition of $\mathcal{G}$ is related to well-separatedness, though it doesn't quite get us there. Then, the collection $\mathcal{K} - \mathcal{G}$ of sets that we have to remove consists of sets C with large $A - C$, i.e. sets whose complement has large intersection with A. But this is exactly those sets for which (2) promises that the corresponding entropy terms are small. So, we get

$$H\left(Y_{\supseteq A} \mid Z_{\mathcal{G}}\right) \leq 2m\varepsilon.$$

We can make two key observations here. First, we didn't actually need to assume that for every sufficiently large $B \subseteq A$, we have $H\left(Y_{\supseteq A} \mid X_B\right)$ small. It would suffice for this to be true of the particular sets that we need based on the relationship between $\mathcal{Y}$ and $\mathcal{Z}$. So, we can hope for much more general condensation properties, that let us adapt to different kinds of latent variable models, giving us something much more generally applicable while still implying meaningful objectivity properties. Second, the development here was guided by the goal of something like a bijection between $\mathcal{J}$ and $\mathcal{K}$, giving pairs of corresponding latent variables. While this is the most obviously compelling objectivity story, sets of variable with somewhat weaker relations may still be unique enough to tell us something interesting about how to understand the joint distribution.

I expect that there is at least some, and possibly a lot, of progress to be made in these directions, which would extend the space of distributions that we can say something interesting about. There are plausibly many unanticipated phenomena here, but I'll give some indication of a few directions to try.

1.2.1. *Almost perfect condensation with a relation.* In the definition of almost perfect condensation, we imposed the condition that $H\left(Y_{\supseteq A} \mid X_B\right) \leq \varepsilon$ when $|A \cap B| \geq r$. This makes some sort of sense, since we can interpret it as saying that X_B has enough elements that are informative about $Y_{\supseteq A}$. But often the cardinality of this set is less closely tied to its informativeness.

We can produce examples by thinking about situations where I has some geometric structure. Suppose Y_A is a latent variable that specifies the approximate shape of some region, and that each observed variable X_i for $i \in I$ is the colour of a point. If we measure many points that are close together, we are less informed about the same than we would be if we measured the colour at a set of points that had the same cardinality, but were more geometrically scattered.

The definition of almost perfect condensation below is wrong, and possibly this even also affects the (m, r, ε) case above. I intend to extract a particular definition and verify that it implies the consequences that I claim here before publishing.

This suggests an idea like the following. Instead of asking about those B such that $|A \cap B| \geq r$, we can ask about those B such that every point of A is within a distance r of some point of $A \cap B$. This leads to the following more general definition of almost perfect condensation.

Definition 2. Let $(X_i)_{i \in I}$ be the random variables of a random variable model and $(Y_A)_{A \in \mathcal{J}}$ the latent variables of an associated latent variable model $\mathcal{Y}$, where $\mathcal{J} \subseteq \mathcal{P}^+ I$. Further, let $\precsim$ be a relation on $\mathcal{P}^+ I$. (Note that $\precsim$ is not assumed to be an order; the idea here is that $A \precsim B$ if B "almost contains" A, in a sense that we are free to choose. For example, we recover the more restrictive sense of almost perfect condensation above if we define $A \precsim B$ to mean that $|B - A| \leq r$.)

We say that $\mathcal{Y}$ is an $(m, \precsim, \varepsilon)$-*almost perfect condensation* if (1—size) $|\mathcal{J}| \leq m$, (2—reconstruction) for all $A \in \mathcal{J}$ and all $B \in \mathcal{P}^+ I$, either $I - B \succsim A$ or $H(Y_{\supseteq A} \mid X_B) \leq \varepsilon$, and (3—Markov) for all $\subseteq$-upward closed sets $\mathcal{F}, \mathcal{G} \subseteq J$, we have $I(Y_{\mathcal{F}}; Y_{\mathcal{G}} \mid Y_{\mathcal{F} \cap \mathcal{G}}) \leq \varepsilon$. Further, we say that $\mathcal{Y}$ is r-*well-separated* if (4) for all $A, B \in \mathcal{J}$, if $|A| \geq r$ then at least one of the sets $A - B$ and $B - A$ has cardinality at least $2r - 1$.

This definition has many of the same good properties as almost perfect condensation, for the same reasons. It is a little more opaque though; it would be easier to understand if we had appropriate examples. The geometric ideas mentioned above seem somewhat promising as a source, though there are some limitative reasons to think that many natural such random variable models, such as joint distributions coming from the colours at different points of a visual scene, will not admit this kind of almost perfect condensation. We will discuss this in the next subsection. However, there may still be *some* useful geometric idea. We can also put other kinds of structure on the set of observed variables. For example, maybe we have many experimental subjects, and we measure many properties of each subject, giving us a set of observations forming a 2-dimensional table. Note here that the different experimental subjects are being treated as giving us different random variables in our joint distribution, rather than being treated as different i.i.d. draws from the same distribution. This could therefore be a better fit for Kolmogorov condensation, which we will discuss in Section §2.

This also doesn't say anything about where $\succsim$ comes from. For a particular choice of $\succsim$, the objectivity theorem tells us something interesting, but it would be more satisfying if the choice of $\succsim$ was in some sense itself approximately unique.

As before, we may do better by adapting ε within a latent variable model, rather than using the same ε everywhere, as is done here for simplicity.

1.2.2. *Medium-scale effects.* The hypothesis of well-separatedness is overly restrictive around what we might think of as medium scale effects, in a sense that we'll look at next. The parameter—either r or $\precsim$—that governs whether latent variables can be reconstructed from observed variables also controls how far apart latent variables must be from each other.

In the case of a visual scene, we might imagine trying to construct a latent variable model using different latent variables for different properties of the objects visible in the scene. For example, we can have a latent with a large contribution set specifying the approximate position and orientation of a tree, with smaller latents filling in details about its shape, e.g. its branches, which are more local and thus have smaller contribution sets. If A and B are the contribution sets of the tree latent and the branch latent, then on the one hand we'd like $I - B \not\gtrsim A$, since if we don't make any measurement of the area with the branch, we won't have a good idea of the approximate shape of the tree. On the other hand, we do want $I - B \gtrsim A$, since we want to rule out the information in the branch latent from being necessary for recovering the tree latent. If we have good separation of scales—e.g., the tree scale is much larger than the branch scale—we can go with the second option, $I - B \gtrsim A$; we don't actually need a sample from the branch to have sufficiently good coverage to infer the large-scale shape. But we'd like to say as much as we can even when we don't have this kind of separation of scales.

While this visual scene idea is described informally, we can come up with particular distributions in which these sort of problems exist, which can be good test cases for refining these ideas. Here I have in mind distributions like those coming from statistical physics, like a random walk or a Ising model near its critical temperature. Whether or not the more specific suggestions above are pointing in a useful direction, it would be good to have any kind of analysis of these distributions. Near the critical temperature, a typical Ising model configuration has a scale-free structure, where it is made up of large domains separated by walls, but then these domains have smaller islands with the same shapes. In order to understand these, physicists introduce coarse-grained variables in renormalization schemes, which are rather like the latent variables that we care about here. One version of this therefore is that we want to know what the objectivity theorem can tell us about the relationship between two different renormalization schemes.

The random walk example is simpler to describe. Given some number N, we will construct a joint distribution on $(X_i)_{i=0}^N$. Let $X_0 = 0$ be a constant, and define $X_{i+1} = X_i + \Delta X_{i+1}$, where each ΔX_{i+1} is an independent Rademacher variable, equally likely to be ± 1. It's tempting to take latent variables that are certain averages of the X_is, but even more simply we can get interesting latent variable models just by making latent variable models into copies of certain X_is. We would like both these kinds of latent variable models to be related by some appropriate objectivity theorem, but we'll discuss the latter kind here for simplicity. Make a binary tree of subintervals of $[1, N]$, where the root is labelled $[1, N]$ and for each node whose interval $[\ell, u]$ has more than one element, its two children are labelled $[\ell, i - 1]$ and $[i + 1, u]$ for some $i \in [\ell, u]$. We'll assume that the tree is reasonably shallow—not too much deeper than $\log_2 n$—but let the tree be otherwise arbitrary. Now, we can construct a latent variable model. Let $\mathcal{J}$ be the set of labels of the tree, and define $(Y_A)_{A \in \mathcal{J}}$ by letting $Y_A = X_i$, where i is the chosen cut point of the interval A, if $|A| > 1$, and otherwise $Y_A = Y_{\{i\}} = X_i$. Because of the Markov property of the random walk, this latent variable model satisfies its own Markov property exactly, whichever tree we use.

Now, for an objectivity theorem, we don't expect any two such trees to correspond exactly; they have different cut points. However, given any variable at layer L of one tree (counting up from $L = 0$ at the leaves), we can be reasonably confident of its value given only variables at layers at least $L - c$, where the exact conditional entropy depends on the choice of c. Instead of a one-to-one correspondence of variable, we have a few-to-one function, where each variable in one tree is determined by a small number of variables in the other tree.

This corresponds to what the objectivity theorem should tell us in the case of almost perfect condensation with a relation. While together with well-separatedness properties we can get a bijection on latent variables, the same ideas give us these few-to-one relations without the well-separatedness assumption. Is this the best we can do? Is this the right way to understand this example? What other class of latent variable models for a random walk does this analysis generalize to? Does it generalize further to the critical Ising model, or to other such examples that we can come up with? It would be good to understand these examples better.

1.3. **Almost perfect condensation and scoring functions.** In Eisenstat (2025) and in Demski (2025), condensation is expressed in terms of some functions, the *simple score* and the *conditioned score*, which relate it to compression. Almost perfect condensation and related ideas are helpful for understanding why the objectivity theorem gives good bounds in certain situations, but the hypotheses of almost perfect condensation don't have a clear relation to compression. It would be interesting if these two perspectives could be unified.

2. Condensation and inductive inference

In Eisenstat (2025), condensation is discussed in a setting where "the given" is a *random variable model*—a joint distribution on a set of variables. This differs from how we may otherwise want to model scientific inference. An alternative model of the given would be a family of observations, that is, the particular observations made, rather than a distribution over possible observations. From the perspective of this model, the idea that the observations can be treated as repeated draws of the "same" set of variables, which are independent and identically distributed, is a theoretical postulate, which may or may not be a desirable way to understand the data. The setting of random variables thus restricts the set of scientific inference situations to which condensation can be applied

However, I expect that the theory of condensation can also tell us similar things when we take the given to be a family of observations, by generalizing the sort of information theory from Shannon information, which applies to probability distributions, to other notions of complexity which apply to individual objects. The best-known of these are the various kinds of Kolmogorov complexity, which are discussed for example in M. Li and Vitányi (2019), with applications to inductive inference treated in more detail in Rathmanner and Hutter (2011).

In this section, we'll discuss research directions involving making *Kolmogorov condensation* work in this alternative setting, and well as some learning-theoretic directions, which might also make sense in the original *Shannon condensation*, but which I think fit more naturally with this idea of observations as individual objects with quantifiable complexity.

2.1. **Kolmogorov condensation.** In order to match the structures of condensation to the setting of inductive inference, we will model our observations as a sequence $(x_t)_{t \in T}$, where the idea is that x_t is the observation that we make at time

t. The set T of all times is a (finite or infinite) initial segment of the natural numbers. Each element x_t will be taken to be a finite string over the binary alphabet $\mathbf{2}$. The broader idea here is that these strings are "computational objects"—they are objects that an agent can act upon with certain operations, and by doing so pose certain hypotheses, which can be a guide to prediction and action.

This differs from the idea of a random variable model, as in Eisenstat (2025). To recall, a random variable model has the form $\left(\Omega, (X_i)_{i\in I}\right)$, where Ω is a probability space, and $(X_i)_{i\in I}$ is a family of random variables on that probability space (satisfying some analytic conditions). We want to replace definitions that use Shannon entropy and mutual information with those using algorithmic-information.

There is a direct analogy here, where the entropy $H(U)$ of a random variable U corresponds to the Kolmogorov complexity $K(u)$ of a string u. Similarly, we can replace relative entropies $H(U \mid V)$ and mutual informations $I(U; V \mid W)$ by algorithmic quantities $K(u \mid v)$ and $I(u; v \mid w)$. In information theory in general, this procedure often takes theorems to theorems, if applied with a little care. For example, we have

$$H(U \mid V) + H(V) = H(U, V) = H(V \mid U) + H(U)$$

and, as discussed by M. Li and Vitányi (2019), there are similar statements satisfied by algorithmic information. For "plain" algorithmic complexity, we have (Theorem 2.8.2 and Corollary 2.8.1—in the fourth edition at least)

$$C(u, v) = C(u) + C(v \mid u) + O(\log C(u, v)),$$

and using a certain variant Kc of conditional prefix complexity, we can do even better (Example 3.8.2):

$$Kc(u, v) = Kc(u) + Kc(v \mid u) + O(1).$$

We can list explicitly the ways that we have to take care here in passing to algorithmic information. Most obviously, we have to add the lower-order correction terms $O(\log(C(u, v)))$ and $O(1)$; the most straightforward analogue with no correction term is not necessarily true. Further, we should understand what is meant by these terms. While these terms bound the dependence of the error on u and on v, there is another parameter here—the choice of the *reference* universal Turing machine in the definition of algorithmic complexity. This is taken to be outside the quantifier of the asymptotic notation, so e.g. the $O(1)$ cannot grow without bound for a fixed reference machine, but it is not bounded with respect to also varying the machine. (If we want to, we might choose a reference machine to make this quantity small.) Finally, we should be careful about expressions with pairs, like $C(u, v)$. In the probabilistic setting, we would use (U, V) to refer to the product random variable. Here, $\langle u, v\rangle$ is a kind of pair data structure, and details like the length of the string $\langle u, v\rangle$ depend on exactly what encoding convention we choose for such pairs.

Following the idea of this analogy, we'd like to give a definition of *latent string models* analogous to latent variable models, and we'd like an analogue of the objectivity theorem to hold. This should be something like the following. A string model is a family $(x_t)_{t\in T}$ as we discussed at the beginning of this section, and an associated latent string model is a pair of a family $(y_j)_{j\in J}$ and a contribution relation $\triangleright \subseteq J \times T$, where there's a distinguished element $\star \in J$, with $\star \triangleright t$ for all t, such that for all $t \in T$ we have $x_t = y_\star(y_j : j \triangleright t)$. In that last equation, we are interpreting the string $y_\star$ as a program—correspondingly the strings in $(y_j : j \triangleright t)$ are "input data". Like with latent variable models, the idea is that each string is determined by the set of latents that contribute to it, but now we are replacing a probabilistic notion of determination by an algorithmic one. Here, we're using an explicit contribution relation, rather than assuming that the contribution relation is set membership, because we want to better control the encoding for the purpose of algorithmic information; if the contribution relation was just set membership, then $(y_j)_{j\in J}$ would become an exponentially long list in which most elements are trivial, which would carry a certain quantity of algorithmic information just in the indices of the nontrivial entries. One could also explore other ways to deal with this though.

Now, following the proof of the objectivity theorem, we should get an algorithmic objectivity theorem for latent string models. Using the notation for working with latent variable models from the original objectivity theorem, this will say something like

$$Kc\left(y_{\supseteq A} \mid z_{\mathcal{F}^\circ}\right) \leq \sum_{B\in\mathcal{F}} Kc\left(y_{\supseteq A} \mid x_B\right) + \sum_{v\in N} I\left(z_{\mathcal{L}(v)}; z_{\mathcal{R}(v)} \mid z_{\mathcal{I}(v)}\right) + \text{error terms},$$

or we could use C and other notions of mutual information if we prefer. (The algorithmic mutual information is here following M. Li and Vitányi (2019), who define

$$\begin{aligned} I_C\left(u : v\right) &= C\left(v\right) - C\left(v \mid u\right) \\ I\left(u : v\right) &= K\left(v\right) - K\left(v \mid u\right) \\ I\left(u ; v\right) &= Kc\left(v\right) - Kc\left(v \mid u\right), \end{aligned}$$

though these conventions are not standard in the field.) This sort of thing would be satisfying, though it would also be good to better understand exactly what the error terms are, and how to keep them reasonably under control. It's possible that this would require some kind of coding tricks.

2.1.1. *The contribution relation in Kolmogorov condensation.* There's at least one thing elided in the attempted proof idea of the algorithmic-information objectivity theorem above: the proof steps that we're considering depend on the contribution relation of the latent string model z. Let's call that relation $\blacktriangleright$, as distinct from the relation $\triangleright$ of y. Taken naively, this will make each of the error terms at least as big as $K(\triangleright)$. In some regime—roughly, when the Kolmogorov complexities of some relevant strings is much larger than the Kolmogorov complexity of the contribution relation—we might regard this as not mattering, so the error terms can be seen as negligible. But we'd like to also have something to say when the complexity of the contribution relations can't be neglected.

The simplest step would be to relativize everything to the relation $\blacktriangleright$. That is, we would have something like

$$Kc\left(y_{\supseteq A} \mid z_{\mathcal{F}^*}, \blacktriangleright\right) \leq \sum_{B\in\mathcal{F}} Kc\left(y_{\supseteq A} \mid x_B, \blacktriangleright\right) + \sum_{v\in N} I\left(z_{\mathcal{L}(v)}; z_{\mathcal{R}(v)} \mid z_{\mathcal{I}(v)}, \blacktriangleright\right) + \text{error terms},$$

which we could combine with

$$\begin{aligned} Kc\left(y_{\supseteq A} \mid z_{\mathcal{F}^*}\right) &\leq Kc\left(y_{\supseteq A}, \blacktriangleright \mid z_{\mathcal{F}^*}\right) + O\left(1\right) \\ &\leq Kc\left(y_{\supseteq A} \mid z_{\mathcal{F}^*}, \blacktriangleright\right) + K\left(\blacktriangleright\right) + O\left(1\right). \end{aligned}$$

Then, only this $K(\blacktriangleright)$ term depends on $\blacktriangleright$, rather than every error term as before.

Losing the whole quantity $K(\blacktriangleright)$ as an additional term seems excessive; we should be able to do even better. One way to think of the above situation is that we're effectively combining the contribution relation $\blacktriangleright$ into the top latent $z_I = z_\star$. We might try various strategies to avoid this, but one option is distribute the information embodied by the contribution relation among the various latents. We could implement this computationally as follows. We've already said that the top latents $y_\star$ and $z_\star$ have to perform the computational roles in their respective models of calculating the given strings x_i from the respective latent strings y_j or z_k. Now, we can further ask them to calculate the contribution relation. Most simply, we can ask that $y_\star(j, i)$ answer the question of whether $j \triangleright i$. By putting the information of the contribution relation into the top latent, this isn't really an improvement over the previous trick. However, now we can generalize this. We can allow $y_\star(j, i)$ to also answer that it has not yet decided whether $j \triangleright i$, and needs more information—information contained in other latents like some y_ℓ with $\ell \in J$. Then, maybe we ask again by computing $y_\star(j, i, y_\ell)$. There's a lot of choices to make and bookkeeping to keep track of here. For example, we need some well-foundedness. This should be something like $y_\star$ having already decided previously that $\ell \triangleright i$, in order to prevent the information in y_ℓ from leaking into x_i when it shouldn't. We need to prevent circularity here. But I hope that by such methods we can prevent the otherwise large overhead associated with the contribution relation.

2.1.2. *Further computational constructions.* Another issue is that if we want to construct a latent string model with many latent strings, each with low information content, discretization errors accumulate. For example, suppose we want many latent strings, each of which we want to think of as an independent coin with a 1% chance of coming up heads. This ought to have low information content, but as a string with a prefix-free encoding, each must take up at least one symbol. We can combine some of them together into larger latent strings, at the cost of inflating their contribution sets, but can we do better by building this into our definitions of string models and latent string models? Could we create something with a hybrid information content, that has both algorithmic and probabilistic aspects?

More broadly, by working out examples it is plausible that other sorts of unnecessary inefficiencies would arise.

2.1.3. *Other complexity notions.* So far, we've discussed Shannon information and variants of Kolmogorov complexity (plain complexity, prefix complexity, the modified conditional prefix complexity Kc). For settings where we're performing inference on observations that can be encoded as strings, we'd like to use complexity notions appropriate to strings, broadly similar to Kolmogorov complexity. However, the Kolmogorov complexities discussed above have many more properties than we really use in the proof of the objectivity theorem. Really, all that we need from a complexity measure is that it (approximately) satisfy certain information-theoretic identities and inequalities, things like $H(u, v) = H(u) + H(v \mid u)$, together with certain string bookkeeping identities like $H(u, v) = H(v, u)$.

There are many different potential ways one could approach defining a kind of complexity for strings. I'll list some that I'm aware of, though others may be attractive for different reasons.

One idea is to use the length of the shortest *straight-line program* as the complexity metric, rather than allowing general computation. There is a lot of work on this that I'm not familiar with, and it has connections to ideas like the widely used Lempel-Ziv compression algorithm, but what I know of it comes from Vanessa Kosoy (Kosoy, 2026). First, in order for straight-line complexity to give an interesting form of condensation, we want to know whether all the necessary information-theoretic properties hold, and with what sort of errors. Next, we can ask about the computational complexity of various operations, most notably finding latent string models with properties like almost-perfect condensation, in cases where they exist. Here, in the case of compressing a single string, there are positive results on efficient computation of close-to-optimal compressions (Rytter, 2003), though there are also negative results (Charikar et al., 2005). This gives some hope that there could be also be efficient computations of interesting latent string models. In any case, optimization over the space of straight-line programs is computable, unlike optimization over all programs for a universal Turing machine.

If we just want a computable variant of algorithmic-information condensation, we can substitute some kind of Kolmogorov-complexity-like measure that also accounts for computation resources, such as by adding a penalty for time or space usage. I expect that there is some version of this such that appropriate algorithmic-information properties hold, allowing us to prove an objectivity theorem. M. Li and Vitányi (2019) have some discussion of Kolmogorov complexity with computational constraints, but I'm not sure if the complexity metrics they define are what we'd need.

Another possibility is to think of complexity using neural nets. Here, the idea is that the complexity of a string would be something like some measure of the complexity of a neural net, plus the surprise of that neural net when asked to predict the string. We can think of the neural nets as trained with local search methods, rather than globally optimal. In such a setting, it's probably too much to ask for the approximate information-theoretic identities to always hold, but we can investigate whether they usually hold for interesting data. For conditional complexity, the analogue of $K(y \mid x)$ should be something like the complexity of a network that is allowed to look at (i.e. has connections in from) a frozen copy of a network trained to predict x well. Will $K_{NN}(x,y) \approx K_{NN}(x) + K_{NN}(y \mid x)$ hold approximately when x and y are two subsets of a pretraining corpus obtained by filtering for different kinds of text, or are input-output pairs of simple algorithmic tasks?

It might also be helpful to consider the situation abstractly. Can we axiomatize some properties that a complexity measure can satisfy, and derive an abstract objectivity inequality? Such ideas may help guide the creation of interesting complexity measure different from those already considered.

2.2. **Online and out-of-distribution generalization.** Here, we'll discuss the application of condensation ideas to get new kinds of control of generalization in learning theory. I'll talk a bit about agency in order to motivate such generalization, but the problems discussed here can also be understood as purely learning-theoretic.

An agent must do many things that they can be confident will work well for them on the basis of their past experience of their world. But the world is not a supervised learning problem, where the challenges one encounters are drawn from the same distribution as those one has dealt with in the past, and where it suffices to do well on average. Agents encounter situations that are meaningfully new, where the right courses of action are underdetermined by their experience. When something new happens, learning theory gives us poor control over whether things that worked in the past continue to work in the future. This shows up in the usefulness of novelty-avoiding methods for avoiding catastrophes, such as in Ebtekar and Cohen (2026). In principal, we'd like to do even more; instead of just avoiding novelty, we'd like to know what kinds of novel actions really are desirable for underlying reasons that we can see as shared with past events, and when we'd like to be more cautious, perhaps by offering human correction, of novel actions. What we want here is to understand the hypotheses and behaviour of agents, rather than treating them as black boxes. The realm of reason, argument, and explanation reemerges; in a certain artificial long run, the best predictors win out by virtue of their track record, and their reasoning can fade into the black boxes of their predictive methods, but in the real world, faced with conflicting plausible advice, one wants more.

We can point out what features of existing learning theory make it insufficient for an understanding of these issues. There are two main kinds of learning theory. First, there is the theory of generalization bounds based on data drawn i.i.d. from a fixed distribution. Here, we can get useful guarantees, but as soon as the distribution changes, we are without adequate guidance from theory. Second, there is online learning theory. Here, we have theorems like various sublinear regret bounds. These control the total number of mistakes, but they don't tell us about the particular novel situations where mistakes are more likely. (We can also consider various kinds of reinforcement learning and active learning; for our purposes here, these have the same issues as online learning, but with further complications.)

A few ideas give success going beyond either of these strands, notably Aaronson, Lee, and J. Li (2026). In this paper, the authors give a PAC learning result across a distributional shift. They show that predictors that depend only on a sparse subset of input features, or more generally on a low-dimensional projection of the input space, can generalize out-of-distribution, as long as the marginal distribution on that quotient space remains approximately constant. This seems complementary to condensation, which gives up interesting low-dimensional subspaces to work with; we'd like to define our subspaces in terms of latent features, rather than only observed features. I'd like to know to what extent we can put these ideas together, and characterize what control the existence of latent variable models with good condensation properties, such as almost perfect condensation, gives us over generalization of a predictor that depends only on a sparse subset of the latent variables. The ideas in Aaronson, Lee, and J. Li (2026) are probably also relevant to the following discussion, though we'll move to a more apparently different language of Kolmogorov condensation.

Consider now the following setting. We have some initial segment $(x_t)_{t=1}^{T_1}$ of data, which is a prefix of a longer segment $(x_t)_{t=1}^{T_2}$ for some $T_2 > T_1$. Suppose that both string models admit latent string models $\left((y_j)_{j\in J}, \triangleright\right)$ and $\left((z_k)_{k\in K}, \blacktriangleright\right)$ with strong condensation properties. For example, this could be the Kolmogorov variant of almost perfect condensation, definition 1. To spell out what this means for $\left((y_j)_{j\in J}, \triangleright\right)$, we can first define the notation $j \sqsupseteq \ell$, for $j, \ell \in J$, to mean that for all $t \in [1, T_1]$, if $\ell \triangleright t$, then also $j \triangleright t$, and also define the shorthand $\{j\triangleright\}$ for the set $\{i : j \triangleright i\}$.

Definition 3. $(y_j)_{j\in J}$ is an (m,r,ε)-*almost perfect condensation* if (1) $|J| \leq m$, (2) for all $j \in J$ and all $A \subseteq I$, if $|\{j\triangleright\} \cap A| > r$, then $Kc\,(y_{\sqsupseteq j} \mid x_A) \leq \varepsilon$, (3) for all $\sqsubseteq$-upward closed sets $\mathcal{F}, \mathcal{G} \subseteq J$, we have $I\,(y_{\mathcal{F}}; y_{\mathcal{G}} \mid y_{\mathcal{F}\cap\mathcal{G}}) \leq \varepsilon$. Further, it is r-*well-separated* if (4) for all $j, \ell \in J$, if $|j\triangleright| > r$ then at least one of the sets $\{j\triangleright\} - \{\ell\triangleright\}$ and $\{\ell\triangleright\} - \{j\triangleright\}$ has cardinality greater than $2r$.

So, we might assume for example that both y and z are well-separated (m,r,ε)-almost perfect condensations, for some sufficiently small m, r, and ε. Note that such condensations need not exist, and even if such a y exists, that doesn't guarantee the existence of z, which must account for the longer segment $(x_t)_{t=1}^{T_2}$.

Under such hypotheses, we'd like to be able to prove something about how a universal learner generalizes from $x_{1:T_1}$ to $x_{(T_1+1):T_2}$, even if this constitutes a distributional shift, or more generally if the sequence $x_{1:T_2}$ is far from i.i.d. The idea here can be stated in terms of the relation between y and z. By the objectivity theorem, we should get some sort of correspondence between the latent strings of y and z, with the exceptions of those latents that contribute to few elements of x, which we can neglect as "noise" rather than "pattern", and of z latents which contribute only to observations with indices in the range $[T_1+1, T_2]$, which may have no y analogue. Now, we can hope that at those $t \in [T_1+1, T_2]$ with no such new "pattern" latents, we can be confident that we have generalized from experience to the best of our abilities. To attempt a more precise statement, we'd like to say that a universal learner might not be confident in a single prediction about what comes next, but their uncertainty should be mostly "noise", in the sense that this question should have low mutual information with other questions that the learner tries to predict, i.e. with other x_t for $t \in [T_1+1, T_2]$. Or in another formulation, all algorithmically simple universal learners ought assign similar probability distributions to x_t.

I hope that such a theory could go some way toward an explanation of why learning algorithms in fact do often generalize well out of distribution. The fact that something like this is true is necessary for the success of human learning, and even more clearly for the success of the methods underlying current AI. Even though we don't care whether chatbots are good at predicting the next token, the unsupervised pretraining objective is extremely useful as a component in creating chatbots that answer question that we do care about. While the fact that it works is so obvious that it's often unacknowledged, this is not promised to us by statistical learning theory!

Such ideas would also provide a starting point for elaboration in various directions. We could get some sort of attribution of knowledge: which previous observations are actually the reason that the learner knows what it knows now? Especially when the underlying things learned are more like values and less like facts, we want to know when what comes next is underdetermined by previous experience.

2.3. **Value learning in toy settings.** I think it would be informative to see how much could be done with such methods in a value-learning setting like the diamond maximizer problem (Yudkowsky, 2015). It is hard to define the amount of diamond in an environment directly from an agent's observations. Instead, one often wants to learn what diamond is from some sort of corpus of examples. Condensation suggest two directions here. On the one hand, as discussed in the section on generalization and learning theory, we can hope to better understand to what extent concepts are really pinned down by given data, and in what realm they become underdetermined and more specification is needed. On the other hand, if an agent's understanding can be seen as a latent string model, then we can hope to speak the agent's language and describe diamond directly. Perhaps we want to use both these methods together, somewhat like what we do when explaining new ideas to each other.

In particular, we want to be able to tell a diamond maximizer a certain amount about what diamonds are, and then have them learn more about the world, while still pursuing diamond. At least for sufficiently tame kinds of ontological revision, we might hope that the broader structure of certain latent strings survives even as our picture of physics and thus of the microstructure of diamonds changes.

3. Condensation and statistical models

In statistics, many different kinds of models with latent variables have been studied. I'm interested in the extent to which condensation can serve as a sort of framework here; when latent variable models[1] are useful for understanding data, can we say that part of the reason that they have good condensation properties? Here, I have in mind models such as Bayesian networks with latent variables, sparse autoencoders, hidden Markov models, and various parametric latent variable models (e.g. mixtures of Gaussians).

Often, we care about *identifiability* of parameters of latent variable models. When parameters are underdetermined by data, it is hard to take them seriously as telling us something meaningful about the world—other parameter values would do just as well. The objectivity theorem gives us both less and more than identifiability.

Strictly speaking, it's in a different setting, because identifiability involves taking a limit as data increases. In Shannon condensation, we use a joint distribution, which we could imagine as being estimated from an increasing sample, but a better analogue, especially in the case of Kolmogorov condensation, is to let the *number* of given strings or given random variables increase. Some details depend on exactly what question we ask, but this will very rarely give us a unique element in the whole space of latent string models or latent variable models. However, we also can get more than identifiability, in that *all*[2] the latent models satisfying certain properties (e.g. almost perfect condensation) will stand in a close relation to each other, not just some parametric or otherwise characterized family. In this way, we can have a confidence in our model without the same dependence on our selection of model class. Though, there are still choices made here, in choosing to be interested in condensation; in choosing a complexity measure; in estimating whether we have found an almost perfect condensation, which is uncomputable if we use genuine Kolmogorov complexity, etc.

While we have been talking about models with latent variables, we should also take note that these ideas can be relevant even to statistical models without latent

[1] *Latent variable models* is here not meant in the specialized sense of Eisenstat (2025), though the senses are close enough that it isn't really reason for confusion.

[2] Here we do mean latent variable model in our technical sense. There's a methodological point here, visible in this word "all"; we must give the term a technical sense if we want to say something about all latent variable models. This is analogous to how e.g. Turing had to give "algorithm" a definition in order to say that *all* algorithms fail to solve the halting problem.

variables. Given a random variable model $(X_i)_{i \in I}$, we can consider latent variable models where the same family $(X_i)_{i \in I}$ is also the family of latent variable, with the same index set I. This is not trivial, because we still have to choose the contribution relation $\triangleright \subseteq I \times I$. There is a real question, for example, of whether there is some choice of $\triangleright$ such that the resulting latent variable model is an almost perfect condensation. In this way, there is a potential to say something about models like Bayesian networks, fully observed Markov models, and maybe some other graphical models, such as those that appear in statistical physics.

3.1. **Bayesian networks.** A Bayesian network (Spirtes, Glymour, and Scheines, 2001) is rather directly a latent variable model. We can thus apply the objectivity theorem, under assumptions such as almost perfect condensation and well-separatedness. In some sense, this tells us that any two Bayesian networks producing the same distribution on the observed variables have to be related to each other in a certain way. This suggests various questions. What does this relation have to do with the actual structure of the graph; can we prove that there is only one possible graph structure? Does knowing something like uniqueness help us actually find latent Bayesian networks with good condensation properties? Are such conditions satisfied in cases where we can actually discover interesting latents from data?

To give more context on the application of the objectivity theorem, we can think about what it says in this case. The Markov condition will hold exactly. So, suppose $(Y_j)_{j \in J}$ and $(Z_k)_{k \in K}$ are the latent variables of two Bayesian networks, each an (m, r, ε)-almost perfect condensation and r-well-separated. By the objectivity theorem, there is a certain bijection, with properties as follows, between certain subsets $S \subseteq J$ and $T \subseteq K$; for simplicity, identify these two subsets along this bijection. Writing the contribution relations of the two models as $\triangleright$ and $\blacktriangleright$, respectively, we have for all $j, k \in S$ that $|\{\triangleright j\} - \{\triangleright k\}| \leq r$ iff $|\{\blacktriangleright j\} - \{\blacktriangleright k\}| \leq r$. Write this relation on S as $\succeq$. Now, we have for all $j \in S$ that $H\left(Y_{\succeq j} \mid Z_{\succeq j}\right) \leq m\varepsilon$ and reciprocally $H\left(Z_{\succeq j} \mid Y_{\succeq j}\right) \leq m\varepsilon$.

3.1.1. *Condensation and Bayesian network graph structures.* I expect that for many distributions, some objectivity property holds in the space of Bayesian networks with latent variables. This would be something like uniqueness for the directed acyclic graph structure in the space of all Bayesian networks which are well-separated almost perfect condensations, and some sort of quantitative approximate uniqueness for the parameters. I haven't tried to prove this, and there may be unforeseen difficulties. In particular, the conclusion of the objectivity theorem only says $H\left(Y_{\succeq j} \mid Z_{\succeq j}\right) \leq m\varepsilon$, whereas it seems like this would be a little easier if we could get $H\left(Y_j \mid Z_j\right) \leq m\varepsilon$.

3.1.2. *Finding latent variable models.* There are a variety of different algorithms that have been studied for the problem of latent causal discovery (Peters, Janzing, and Schölkopf, 2017). One could ask for progress either from Bayesian networks to condensation or vice versa here. On the one hand, it would be good if some algorithms for latent causal discovery could be applied more generally to find other kinds of latent variable models. Maybe if we assume that latent variable models with good condensation properties exist, we could give conditions under which existing latent causal discovery algorithms could find them. On the other hand, we could try to create new algorithms for discovering latent variable models, taking inspiration from hypothesis involving condensation properties. Such algorithms could then in particular be applied to find Bayesian networks with latent variables.

3.1.3. *Condensation as a reason for Bayesian networks.* Some latent variables constructed using latent causal discovery algorithms have been seen to make sense, and to give us better understandings of the corresponding problem domains (Xie et al., 2024). Generally in such cases, we have reason to think that the Bayesian networks are approximately unique based on identifiability results. I conjecture that these Bayesian networks are also approximately unique in the sense of the theory of condensation, and that this uniqueness can be justified by the objectivity theorem.

3.1.4. *Algorithmic-information condensation.* So far, we've been using the random-variable form of condensation, which seems like a more straightforward counterpart of Bayesian networks. However, it's natural to also consider the data as strings; rather than a joint distribution on the observable variables, we can start with actual observations of observed variables. To prove an objectivity theorem here, we'd probably need something beyond almost perfect condensation; rather than reconstructing latent string from arbitrary sets of observations of sufficient size, we'd want to reconstruct them from sets involving many observations of the *same* variable, or something along such lines. Such an approach could help with the analysis of more general Bayesian network models with repeated elements, such as hidden Markov models or plate models.

3.2. **Sparse autoencoders.** Overcomplete dictionary-learning methods are interesting for their ability to find meaningful structure in high-dimensional spaces. Recently, there has been interest in applying such ideas to neural networks, in the form of sparse autoencoders, or SAEs (Cunningham et al., 2023). Here, we can consider a simple toy model. Fix some high-dimensional real vector space $\mathbb{R}^d$ and take n "dictionary" vectors $(g_j)_{j=1}^n$ in this space. Then, we can construct a dataset of sparse combination of these vectors. To construct one such vector, pick s-many indices in the set $\{1, \ldots, n\}$ uniformly at random, for some fixed $s \ll n$, and sum the corresponding vectors with standard normal coefficients. To put this another way, we can think of this as constructing a family of coefficients $(\alpha_{i,j})_{j=1}^n$, where s-many of the coefficients $\alpha_{i,j}$ are drawn from a standard normal, and the rest are zero, and then forming an observed vector $x_i = \sum_{j=1}^n \alpha_{i,j} g_j$. We can do this independently m times to get a dataset $(x_i)_{i=1}^m$. While these elements are vectors in $\mathbb{R}^d$, we can discretize them to get a representation with finite algorithmic information.

With this toy model, we can ask whether a latent string model with good condensation properties exist. I conjecture that in an appropriate regime of (n, m, s), and if the dictionary vectors are close to pairwise orthogonal, we will have an almost perfect condensation, given in a natural way by the dictionary vectors. Further, we can vary these conditions. What if the vectors are not close to orthogonal? What if there are more interesting correlations in the probabilities of cooccurrence of the different dictionary elements? Such phenomena have been observed in SAEs derived from real data, and I hope that condensation can help us find better ways to understand data like this.

4. Language and logic

Suppose we have two agents who make observations represented as families of strings $(x_i)_{i=1}^n$ and $(y_j)_{j=1}^m$. If we assume that there are latent string models for these respective families, which the agents find and use, and that there is also such a latent string model for the union of the two families, then we may expect the different latent strings that the agents use to be related. But how should the agents actually talk to each other? How can they establish propositions, which they do not already know the truth of, but which they can both assign the same meaning to, and which they can thus e.g. ask each other about? Where does the combinatoriality of language enter? Where are logical connectives, entities, relations, and quantifiers? The use of general computable operations on strings gives us some sort of combinatoriality here, but the full story has yet to be discovered.

5. Further directions

There's lots of cool stuff that we might hope for condensation to be able to do. In most cases, I don't believe that condensation can solve the problem alone, but I think it might be able to find some interesting structure and shed light on what's still needed. But regardless, I'll speculate a little here.

5.1. **Emergence.** We can produce many random variable models that we want to describe as having emergent or higher-level phenomena within them. For example, the basic variables could be the spacetime points of a stochastic PDE or a cellular automaton, or they could be the low-level degrees of freedom of a many-body system in equilibrium, maybe especially one that exhibits critical phenomena. It's then very easy to hope that emergent entities, things like wavepackets and solitons, vortices, gliders, and domain walls, would be latent variables of latent variable models with good condensation[3] properties. Kolmogorov condensation may be especially helpful here, since it may be easier to deal with the contribution relation for a small set of wavepackets, rather than dealing with latent variables for all possible wavepackets, most of which are not present in a given field configuration.

I am optimistic about this working, but it must decided by actual examples.

I'd also like to know to what extent we can see the resulting latent variables as simultaneously telling us about different levels of description. For example, can we have latents both for something like particle trajectories, and for corresponding macroscopic variables like local pressure in a region? How do different levels of description interact in this language?

5.1.1. *Emergence of agency, telos.* Continuing in this direction, we can ask to analyze biological, purposive, or agentic phenomena. Here, it's easy to want to tell a story where this all works very well and entities like agents, beliefs, and intentions appear as latents, but that's not what I expect. For example, I don't think that the nature of representation, with its interplay of mind-to-world and world-to-mind directions of fit, is something that condensation can adequately deal with by looking for a decomposition into approximately conditionally independent parts.

To the extent that we are limited in this way, I'd instead want to see what condensation can give us, and what's left afterward. A biological system will have purposive aspects, but it's also made of physical stuff, and it will display the kinds of

[3]Admittedly the name condensation can become a source of awkwardness here.

physical emergence discussed above. We might expect to identify some interesting latent variables corresponding to some emergent entities, but also for this to leave a residuum. Then, we wouldn't get latent variable models with good condensation properties, since these are global objects. Thus, I'd want a notion of something like a latent variable model that partially has good condensation properties, and is therefore partially approximately unique.

In any case, I think that any sort of result—positive or negative—that tells us how condensation ideas relate to purposive systems would be extremely interesting.

5.2. **Decision and action.**

5.2.1. *Decision theory and counterfactuals.* Theories like causal decision theory (Weirich, 2024), timeless decision theory (Yudkowsky, 2010), and functional decision theory (Yudkowsky and Soares, 2018) rely on a notion of *subjunctive conditionals*, or *counterfactuals*, in order to prescribe action: *what would happen if I did such and such*? This is discussed for example in Section 5 of Yudkowsky and Soares (2018), where the representations provided by causal graphical models and the definition of counterfactuals in terms of them from Pearl (1997) are suggested as a model of the sort of counterfactuals necessary here.

The problem that this raises, and the subject of the substantive disagreement between the causal decision theorist and the timeless decision theorist, is which counterfactual notion is appropriate for use by an agent choosing between actions. Alternatively, we could say this a different way: if we define $A \mathrel{\Box\!\!\rightarrow} B$ to mean that B would be true, counterfactually given A, in the sense appropriate for agents considering actions, then when is $A \mathrel{\Box\!\!\rightarrow} B$ true? Is it determined by whether A causes B in a sense coming from physics? If so, how can we tell from a description of a physical theory what counterfactual sentences it endorses, even a physical theory in which causality becomes more problematic, as in Maxwell's electromagnetism, as discussed in Chapter 7 of Frisch (2014), or in quantum mechanics? Or is it determined instead in a way that involves considering an action of an agent as a mathematical fact about an abstract decision procedure, as Yudkowsky and Soares advocate? Does this involve such a general notion of logical counterfactual as to promise that there is a fact of the matter about whether $\neg\mathrm{TPC} \mathrel{\Box\!\!\rightarrow} R$, where TPC is the twin prime conjecture, and R is the Riemann hypothesis?

Condensation is generally motivated by a desire to discuss scientific theories as not only predictive, but also explanatory. Within this, one additional desideratum that we can ask of a theory, beyond its being predictive, is that it also answer counterfactual questions. We have examples in the sciences of theories that answer counterfactual questions, though not always as satisfyingly as we'd like.

Theories that take the form of differential equations with existence and uniqueness of solutions answer counterfactual questions (i.e. Cauchy problems), though there are often subtleties, including around boundary conditions and around statistical-mechanical reasoning, as discussed by Frisch (2014). Further, for our purposes here, an *action counterfactual*, of the form needed in decision theory, is an event on a different level of description, which may not straightforwardly translate into a Cauchy problem.

In another direction, theories that take the form of causal graphical models answer counterfactual questions. Here, we are again constrained in which events provide the appropriate "initial conditions" for a meaningful counterfactual question. An agent hoping to ask a causal graphical model about the effects of an action must hope that this action corresponds to one of the causal variables of the model! For example, like with differential equations, we might be at a loss as to what to do if the causal variables are more "low-level" than the action. A class of theories related to causal graphical models is factored space models (Garrabrant et al., 2024). In this context, Garrabrant (2021) defines a notion of *counterfactability*, so that which events we can make sense of counterfactuals with respect to is a derived property of the model, rather than being constrained to those events that are set up as assignments to causal variables as in causal graphical models.

I'd like to know whether condensation can help us understand why scientific theories, which we might have expected to be merely predictive, turn out to also give us notions of counterfactuals, like differential equations and causal graphical models do. Further, can we understand why events that we would describe as actions turn out to be counterfactable?

While latent variable models are in some ways rather like causal graphical models, they don't have a correspondingly well-developed theory of counterfactuals. Upward cones of latent variables are more natural objects than the latent variables themselves, and in particular the objectivity theorem is stated in terms of upward cones of variables, so we won't a priori have good objectivity around a counterfactual on a latent variable without further specifying counterfactual values for its whole upward cone. However, this situation resembles the notion of history in factored space models, and so one suspects that this is the same problem that Garrabrant (2021) is addressing with the notion of counterfactability. Can we give a notion of counterfactability for latent variable models, and will it satisfy an objectivity theorem? Possibly this would require stronger hypotheses than the existing objectivity theorem, but maybe this is what we would need to do in order to understand why agents' actions are counterfactable.

We critiqued the ability of differential equations and causal graphical models to determine action counterfactuals on the grounds that a given model may be too "low-level" to make sense of the proposed action. However, the discussion above on emergence gives some hope of "low-level" and "high-level" events living together in appropriate latent variable models. It would thus be interesting to further combine these ideas with counterfactuals.

While casual graphical models and factored space models work in something like an i.i.d. setting, this is not the actual situation of an agent who is confronted with different possibilities of action to choose between. Instead, we might want a counterfactual theory in which events "only happen once". Kolomogorov condensation may be helpful for bridging this gap.

5.2.2. *Functional decision theory.* Yudkowsky and Soares (2018) develop functional decision theory based on the hypothesis that there is a kind of subjunctive dependence similar but not identical to physical causation. Can we find something like this using condensation? Even if their subjunctive dependence is in some sense natural, we expect physical causation to also be natural. Thus, we might have two different natural counterfactual notions coexisting. One way this could happen is if we have a single counterfactual relation, but we have some sort of physical variables and some logical variables coexisting, so that there is both a physical and a logical counterfactual for a given action, understood in two different ways, as two different latent variables. To the extent that we can use condensation to make sense of the idea of high-level and low-level causation coexisting, that might also provide a model for how this might work.

5.2.3. *Cooperation and Schelling points.* When consider decisions involving multiple agents, as in game theory, it is standard to take for granted that the agents share a priori a basic understanding of the situation. They understand that there are various agents, that they are playing a game, and what the game is. They may have different knowledge, but this knowledge is represented probabilistically within an a priori shared probability space. These assumptions are the reasons that strategic behaviour, such as playing Nash equilibria, makes sense. One expects that players will try to understand each other's behaviour, and pursue their own ends.

However, some phenomena seem to ask us to reexamine this assumption, and more explicitly account for players' understanding of each other. One source of this is cooperation. While agents may establish mechanisms such as reputation for incentivizing cooperative behaviour, some authors such as Rapoport (1966) have argued that agents should sometimes be able to cooperate even without this sort of incentive, based on reflexive reasoning about themselves and each other. There are some formal models of this, such as Barasz et al. (2021), which uses the modal logic of provability to model this sort of reasoning. However, I think this is still incomplete as an explanation of how such cooperative behaviour can occur. Can we use ideas from condensation to describe how agents might reason about each other in playing cooperative games? In particular, agents need to understand the relation between one's own thinking and the understanding on the part of another of that thinking. Can we come up with latent variables making that connection?

These issues become more complicated when we consider that in general there is not a single obvious cooperative strategy for players to coordinate on, but rather they must collectively decide on a cooperative strategy, which involves bargaining since players interests can conflict to a degree. Even when there is not conflict, players who need to coordinate with limited communication must find focal points to coordinate upon. These ideas are discussed by Schelling (1960). Here, we need to understand how agents represent the game and each other in order to understand their behaviour, and we may be able to make progress on this with ideas from condensation.

5.3. **Organization of scientific theories.** Much of what makes science what it is takes place not on the level of knowledge of facts, but on a more abstract level of what we might think of as understanding, or as the organization of scientific theories. Some of the smaller steps from knowledge of facts to understanding can be see in some kinds of statistical practice, as we discussed in Section §3. But as theories develop, they often acquire mathematical and conceptual aspects that can't obviously be expressed in terms of latent variables in this way.

We can try to apply condensation methods. For example, we can take as given strings of a string model either scientific observations, or statements made by scientists, or scientific texts, or solutions to scientific problems. Then, we can hope for latents strings which sometimes express facts, but also maybe express broader concepts, ideas, principles, or problem-solving strategies, which we feel better capture our understanding of the theory. We might expect that the solutions to problems are determined by facts and concepts like this, together with idiosyncratic residual details, as is required by the definition of latent string models. We also might expect that both facts and deeper concepts can be reconstructed from examples of solutions to problems that draw on these concepts. The Markov condition is more problematic; we expect of a mature science that its concepts are deeply related, that it holds together in a way beyond a collection of separately contingent brute facts.

Since I think this sort of interrelatedness is related to the mathematical development of a theory, it's natural to consider especially mathematical examples. We could think of the given strings as labelling truth and falsity[4] of, say, short sentences in the language of arithmetic; or as proofs and exercises from a book developing the foundations of some mathematical theory, along the lines of Euclid or Bourbaki; or as the proofs from a library of formally verified mathematics. To what extent is there a latent string model with good condensation properties? Can we understand the resulting strings as things like ideas and proof strategies? How comprehensible are they to humans; to what degree to we recognize them as corresponding to our own organization of the subject matter?

One issue that arises is that it is possible to answer the same question in many ways, each using different concepts. A choice of a latent string model involves picking one of these as the one described by the contribution relation. Do we want to generalize the notion of latent string model to allow different such answers to coexist?

In general, I'd like to know how well this works and what sort of concepts can be exhibited as latents in this way. I do expect, similarly to what was said in Section §5.1.1, that there will not exist latent string models with global good condensation properties. This has to do with the interrelations of different concepts as mentioned above in the context of the Markov property. For example, suppose that the theoretic statements that we want to organize pertain to the solubility of *Diophantine equations*—polynomial equations for which we want integer solutions, such as $x^3 + y^3 = z^3$. We can fill books with ideas, but they don't seem to stand in conditional independence relations. Say we reduce the equation modulo p for different primes p. To some extent the resulting analyses will be "independent" with respect to solving the problem; it is sufficient to show that no solution exists modulo p for a single prime p, and then no solution can exist to the original equation. But precisely this suggest that they are not probabilistically independent, since an obstruction modulo one prime is evidence that there may be obstruction modulo others. And further concepts, such as quadratic reciprocity, relate analyses modulo different primes to each other is more complicated ways. Overall, it's not clear what would really constitute a piece of independent information pertaining to some primes and not others. Or suppose instead that we identify different strategies, such as reduction modulo p, changes of variables, infinite descent, real-variable asymptotics, etc. Often there are many different ways to solve a particular Diophantine equation, using different such strategies, and using the strategies and having them

[4]Here, I'm thinking of some of the ideas about Kolmogorov condensation sketched in Section §2, where the idea is that the statement is the input to some program, which must output correctly whether it is true or false. There may turn out to be problems with making the complexity here a single bit—truth or falsity—so it may be more convenient to work in a setting where the strings of the string model carry more information. This shouldn't change too much of the discussion here.

interact in different ways. These interactions aren't really a simple combining of different pieces of information, but are mediated by further choices, like choices of functional forms, algebraic manipulations, or inductive hypotheses.

To the extent that strong global condensation properties cannot be achieved, I'd like to better understand what condensation can tell us about the organization of such theories. Can we have locally good condensations, which give objective constraint to some aspects of a latent string model but not others? I'd like to better understand what makes the sort of organization that we conduct when we talk about the concepts belong to such a subject constitute understanding, even if it is not objective in the same way as almost perfect condensation. I hope that better understanding that part of the organization that is more objective can get us closer to this.

Alternatively, if we want to be very optimistic about the power of condensation to organize mathematical understanding, we can ask for a description of what kind of conceptualization leads us to believe in the consistency of higher theories of arithmetic and set theory, such as theories invoking strong large cardinal axioms.

Aside from the mathematical sciences, music also stands out as a domain where independent brute facts quickly give way to concepts standing in more complex interrelations. Any such domain would be an interesting setting for providing difficult cases for the human comprehensibility of concepts that appear as latents, and for understanding the limits of the existence of good condensations.

5.4. **Conceptual revision.** While we might want to think of the understanding in a moment as organized in terms of concepts, the understanding changes over time. Further, concepts are not just added, but rather the whole understanding changes in a more holistic way. This *conceptual revision* can be see in the history of science, as in Kuhn (1962) and Kuhn and Mladenovic (2024), and in the development of an individual.

Conceptual revision is particularly interesting in connection with the demands that we make on theories to explain and not just to predict. For example, if an agent express their values in terms of certain concepts, this leaves them with a question regarding how to continue to value those things that they value after such a revision. If agents talk to each other in a language that is suited to a certain conceptual structure, then they must decide how to go on talking after revision. If an agent understood an experience that they had in the past in terms of certain concepts, then they must decide how to learn from and more generally make use of that experience once they later begin to think in a fundamentally different way.

Of course, when human beings undergo conceptual revision, we are often still able to deal with the things that were the things of our surroundings beforehand, and we are able to go on valuing what we previously valued. What the things of our surroundings are, and what the things of value are, has changed for us, but we can look at the situation from a diachronic overview and see a continuity between the things before and the things afterward. To be clear, this can be hard, and we don't necessarily succeed, but we have some basic ability to do this, and to some extent doing so is necessary for someone to continue to be themself.

It's not clear that condensation can talk about conceptual revision. Rather, it seems to give us conditions under which there's no conceptual revision to do. But, as we discussed above, we may instead choose to understand latent models that only locally have good condensation properties. Then, the remaining irresolvable part may move around, perhaps settling down into good condensation properties with sufficient accumulation of evidence, or perhaps only leaving us with a new set of choices to make. Here, we may see the possibility of conceptual revision. I would be very happy to have anything like this.

References

Aaronson, Scott, Lin Lin Lee, and Jiawei Li (May 12, 2026). *Sparsity and Out-of-Distribution Generalization*. arXiv: `2603.07388[cs.LG]`.

Barasz, Mihaly et al. (Apr. 5, 2021). *Robust Cooperation in the Prisoner's Dilemma: Program Equilibrium via Provability Logic*. arXiv: `1401.5577[cs.GT]`.

Charikar, M. et al. (July 2005). "The smallest grammar problem." In: *IEEE Transactions on Information Theory* 51.7, pp. 2554–2576.

Cunningham, Hoagy et al. (Oct. 4, 2023). *Sparse Autoencoders Find Highly Interpretable Features in Language Models*. URL: `http://arxiv.org/abs/2309.08600`.

Demski, Abram (Nov. 9, 2025). *Condensation*. LessWrong. URL: `https://www.lesswrong.com/posts/BstHXPgQyfeNnLjjp/condensation`.

– (Jan. 23, 2026). *Condensation & Relevance*. LessWrong. URL: `https://www.lesswrong.com/posts/2x9yatKKTRMabQAWq/condensation-and-relevance`.

Ebtekar, Aram and Michael K. Cohen (Apr. 15, 2026). *Golden Handcuffs make safer AI agents*. arXiv: `2604.13609[cs.LG]`.

Eisenstat, Sam (2025). "Condensation: a theory of concepts." In: *ODYSSEY 2025 Conference*.

Frisch, Mathias (2014). *Causal Reasoning in Physics*. Cambridge: Cambridge University Press.

Garrabrant, Scott (Sept. 23, 2021). *Temporal Inference with Finite Factored Sets*. arXiv: `2109.11513[cs.AI]`.

Garrabrant, Scott et al. (Dec. 20, 2024). *Factored space models: Towards causality between levels of abstraction*. arXiv: `2412.02579[cs]`.

Gillen, Jeremy and Daniel Chiang (Mar. 4, 2026). *A summary of Condensation and its relation to Natural Latents*. LessWrong. URL: `https://www.lesswrong.com/posts/agw7HhW4cWjADpBgo/a-summary-of-condensation-and-its-relation-to-natural`.

Kirchner, Jan (Apr. 8, 2026). *Elementary Condensation*. LessWrong. URL: `https://www.lesswrong.com/posts/JKexLeze9kNhCYYzT/elementary-condensation`.

Kosoy, Vanessa (Mar. 27, 2026). *Stringological sequence prediction I: efficient algorithms for predicting highly repetitive sequences*. arXiv: `2603.26852[cs.FL]`.

Kuhn, Thomas S. (1962). *The structure of scientific revolutions*. Chicago: University of Chicago Press.

Kuhn, Thomas S. and Bojana Mladenovic, eds. (May 2024). *The Last Writings of Thomas S. Kuhn: Incommensurability in Science*. Chicago, IL: University of Chicago Press.

Li, Ming and Paul Vitányi (2019). *An Introduction to Kolmogorov Complexity and Its Applications*. Texts in Computer Science. Cham: Springer International Publishing.

Pearl, Judea (1997). "Causation, Action, and Counterfactuals." In: *Logic and Scientific Methods: Volume One of the Tenth International Congress of Logic, Methodology and Philosophy of Science, Florence, August 1995*. Ed. by Maria Luisa Dalla Chiara et al. Dordrecht: Springer Netherlands, pp. 355–375.

Peters, Jonas, Dominik Janzing, and Bernhard Schölkopf (2017). *Elements of Causal Inference: Foundations and Learning Algorithms*. The MIT Press.

Rapoport, Anatol (1966). *Two-person game theory; the essential ideas*. Ann Arbor, University of Michigan Press.

Rathmanner, Samuel and Marcus Hutter (June 3, 2011). "A Philosophical Treatise of Universal Induction." In: *Entropy* 13.6, pp. 1076–1136. arXiv: `1105.5721[cs.LG]`.

Rytter, Wojciech (June 13, 2003). "Application of Lempel–Ziv factorization to the approximation of grammar-based compression." In: *Theoretical Computer Science* 302.1, pp. 211–222.

Schelling, Thomas C. (1960). *The strategy of conflict*. Cambridge, Harvard University Press.

Spirtes, Peter, Clark Glymour, and Richard Scheines (Jan. 29, 2001). *Causation, Prediction, and Search*. The MIT Press.

Weirich, Paul (2024). "Causal Decision Theory." In: *The Stanford Encyclopedia of Philosophy*. Ed. by Edward N. Zalta and Uri Nodelman. Winter 2024. Metaphysics Research Lab, Stanford University. URL: `https://plato.stanford.edu/archives/win2024/entries/decision-causal/`.

Xie, Feng et al. (2024). "Generalized Independent Noise Condition for Estimating Causal Structure with Latent Variables." In: *Journal of Machine Learning Research* 25.191, pp. 1–61.

Yudkowsky, Eliezer (2010). *Timeless Decision Theory*. San Francisco, CA. URL: `http://intelligence.org/files/TDT.pdf`.

– (Dec. 18, 2015). *Diamond maximizer — LessWrong*. URL: `https://www.lesswrong.com/w/diamond-maximizer`.

Yudkowsky, Eliezer and Nate Soares (May 22, 2018). *Functional Decision Theory: A New Theory of Instrumental Rationality*.